Observation Theory

Geometry, Distortion, and Reliability as Properties of the Observer

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Abstract

We propose *Observation Theory*: geometry, distortion, and reliability are properties of the *observation*—the consumer's read operator, budget, and channel—not of the object observed. We derive, from a second-order expansion of the downstream metric, that the operative distortion of a lossy code is the reconstruction error read through the consumer, $d_{\mathcal{O}} = \mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta)$, where Σ_δ is the error covariance and $P_{\mathcal{C}} = \mathbb{E}[J^\top G\, J]$ the Fisher-metric-weighted sensitivity of the consumer. Reconstruction distortion is exactly the corner $P_{\mathcal{C}} = I$; there Shannon rate–distortion and Gauss least-squares live. We work the theory out from first principles: the read operator is derived for softmax attention (query second moment on the across-token–demeaned error), for a gradient-descent optimizer (the loss Hessian, *exactly*), and for retrieval; the read operator is shown recoverable from the consumer alone by a Jacobian second moment; the retrieval capacity threshold is derived from extreme-value theory as $\mu_{\mathrm{crit}} \sim \sqrt{2 \ln N}$; and a consumer-relative Gaussian rate–distortion function is obtained by reverse water-filling on the $P_{\mathcal{C}}$-weighted source spectrum, reducing to Shannon at $P_{\mathcal{C}} = I$. This yields a more general form of the source-coding bound: a consumer-lossless floor $R_{\mathcal{C}}(0) = H(\Pi X) \le H(X)$, so one can be *lossless to the consumer* while compressing below the source entropy—the entropy of the read subspace, not the source, being the true floor; Shannon is the full-read corner $\ker P_{\mathcal{C}} = \{0\}$, and the information bottleneck the non-geometric generalization. Each derived face is then tested against a sealed, before-run prediction: distortion (`demonstrated`, `replicated`, out-of-sample), identifiability (16× chance, blind), capacity (death located to 6%), and rate (a fixed-budget verdict *inverts* under budget-matched observation on real embedding manifolds). A fifth conjecture—a density quotient restoring the invariant in non-spectral retrieval—is *refuted* across four prospective tests and carried as an honest negative. Classical rate–distortion is not wrong; it is the picture one sees after blurring out who is reading.

1 Introduction

Shannon fixed a distortion measure—mean-squared error, the same in every direction—and asked for the fewest bits that meet it [1, 2]. Gauss, earlier, fixed the same measure and asked for the estimate that minimizes the sum of squared residuals [3]. This paper argues that fixing the measure is the error. The distortion that matters is not the error a code makes but the error a consumer *reads*, and different consumers read different subspaces of the same error. Once the distortion is the one the downstream task induces, reconstruction is a single corner of a larger theory—the corner where the consumer is assumed to weight all coordinates equally.

We call this frame *Observation Theory* and develop it in two registers. The first is analytic: from a second-order expansion (§2) we obtain a single controlling quantity, $\mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta)$, and derive its instances—the read operator for softmax attention and for a gradient optimizer (§3), its recoverability from the consumer alone (§5), the capacity threshold of single-stage observation (§6), and a consumer-relative rate–distortion function (§8). The second register is empirical and adversarial to ourselves: each derived claim was written down and cryptographically sealed *before* the measurement that tests it, bars fixed ex ante, misses recorded as misses. Four faces hold; a fifth is refuted (§9); eleven negatives stand with the same prominence as the positives (§10). The recurring demonstration is a single lossy code whose *downstream verdict inverts* when nothing changes but the consumer.

2 The controlling quantity

Definition 1 (Instance). *An instance is $(X, \mathcal{C}, \mathcal{Q})$: a representation X with vectors $x \in \mathbb{R}^D$, a* consumer *$\mathcal{C} : \mathbb{R}^D \to \mathbb{R}^m$ with a divergence D_Y on its outputs, and a family $\mathcal{Q}$ of lossy codes producing $\hat{x} = x + \delta$. The downstream distortion is $d_{\mathcal{O}} := D_Y(\mathcal{C}(x), \mathcal{C}(\hat{x}))$, measured on the consumer output, never assumed from the ambient metric of X.*

Proposition 1 (Consumer-projected distortion). *Let D_Y be twice differentiable with $D_Y(u, u) = 0$ and $\nabla_v D_Y(u, v)|_{v=u} = 0$, and let $G(u) = \nabla_v^2 D_Y(u, v)|_{v=u} \succeq 0$ be its output information metric. Let $J(x) = \partial\mathcal{C}/\partial x$. For a code with error $\delta = \hat{x} - x$ whose conditional covariance given x is $\Sigma_\delta(x)$,*

$$d_{\mathcal{O}} \;=\; \tfrac{1}{2}\,\mathrm{tr}\big(P_{\mathcal{C}}\,\Sigma_\delta\big) \;+\; o(\|\delta\|^2), \qquad P_{\mathcal{C}} \;:=\; \mathbb{E}_x\big[\,J(x)^\top G(\mathcal{C}(x))\,J(x)\,\big], \quad \Sigma_\delta = \mathbb{E}_x[\Sigma_\delta(x)],$$

provided $\Sigma_\delta(x)$ is (conditionally) uncorrelated with the point-dependent weight. Reconstruction distortion $\mathbb{E}\|\delta\|^2 = \mathrm{tr}(\Sigma_\delta)$ is the case $P_{\mathcal{C}} = I$ ($G = I$, $J = I$).

Proof. Expand $\mathcal{C}(x+\delta) = \mathcal{C}(x) + J(x)\delta + O(\|\delta\|^2)$ and $D_Y(\mathcal{C}(x), \cdot)$ to second order about its minimum. The value and gradient terms vanish by $D_Y(u, u) = 0$, $\nabla_v D_Y|_{v=u} = 0$, leaving $D_Y(\mathcal{C}(x), \mathcal{C}(x+\delta)) = \frac{1}{2}(J\delta)^\top G(J\delta) + o(\|\delta\|^2) = \frac{1}{2}\,\delta^\top\big(J^\top GJ\big)\delta + o(\|\delta\|^2)$. Take $\mathbb{E}_x$; under the stated uncorrelatedness $\mathbb{E}_x[\delta^\top(J^\top GJ)\delta] = \mathrm{tr}\big(\mathbb{E}_x[J^\top GJ]\,\mathbb{E}_x[\Sigma_\delta(x)]\big) = \mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta)$. □

Proposition 1 is the theory in one line: *the distortion that governs a task is the reconstruction-error covariance projected onto the consumer's Fisher-weighted sensitivity $P_{\mathcal{C}}$.* The rest of the paper derives $P_{\mathcal{C}}$ for concrete consumers, shows it is recoverable, and studies what the observer's channel and budget do to it. (We drop the harmless factor $\frac{1}{2}$.)

3 The read operator, derived

Proposition 2 (Softmax attention). *Let a consumer read keys $\{k_s\}_{s=1}^S$ through a query bank: logits $l_s = q^\top k_s/\sqrt{D}$, output $p = \mathrm{softmax}(l)$, divergence $D_Y = \mathrm{KL}$. Under a key perturbation δ_s,*

$$d_{\mathcal{O}} \;=\; \tfrac{1}{2D}\,\mathbb{E}_q\,\mathrm{Var}_s^p\big(q^\top\delta_s\big) \;+\; O(\|\delta\|^3) \;=\; \tfrac{1}{2D}\,\mathrm{tr}\big(Q\,\bar{\Sigma}_\delta\big) + O(\|\delta\|^3),$$

where $Q = \mathbb{E}_q[qq^\top]$ is the query second moment and $\bar{\Sigma}_\delta = \sum_s p_s\,\tilde{\delta}_s\tilde{\delta}_s^\top$ is the p-weighted covariance of the across-token–demeaned *errors $\tilde{\delta}_s = \delta_s - \sum_{s'} p_{s'}\delta_{s'}$.*

Proof. The Fisher metric of softmax is $G = \mathrm{diag}(p) - pp^\top$, so $\mathrm{KL}(p\|\mathrm{softmax}(l+\Delta l)) = \frac{1}{2}\Delta l^\top(\mathrm{diag}(p) - pp^\top)\Delta l + O(\Delta l^3) = \frac{1}{2}\mathrm{Var}_s^p(\Delta l_s)$ with $\Delta l_s = q^\top\delta_s/\sqrt{D}$. The $-pp^\top$ term is exactly the softmax's invariance to a common additive logit shift; it removes the p-mean of δ_s, yielding the demeaned $\tilde{\delta}_s$. Averaging $q^\top\tilde{\delta}_s\tilde{\delta}_s^\top q$ over q gives $\mathrm{tr}(Q\,\bar{\Sigma}_\delta)$. □

Two consequences are exactly the phenomena we later measure. First, $P_{\mathcal{C}} \propto Q$: an *isotropic* query bank gives $P_{\mathcal{C}} \propto I$ (reconstruction-like), while a subspace-concentrated bank projects onto that subspace—so the read operator, and hence the winning code, changes with the query distribution. Second, only the *demeaned* (tangential) part of the error survives; the across-token mean is a nuisance the softmax cannot see.

Proposition 3 (Gradient-descent optimizer; exact)**.** *Let $L(\theta) = \frac{1}{2}\theta^\top H\theta$ with $H \succeq 0$, gradient $g = H\theta$, and a compressed gradient $\hat{g} = g + \delta$ used in one step of size η. Then*

$$L(\theta - \eta\hat{g}) - L(\theta - \eta g) = \tfrac{1}{2}\eta^2\, \delta^\top H\delta - \eta\, \delta^\top H(\theta - \eta g),$$

whose error-quadratic part is $\frac{1}{2}\eta^2 \operatorname{tr}(H\, \delta\delta^\top)$; averaging the linear cross term over zero-mean, direction-uncorrelated error gives $\mathbb{E}[d_{\mathcal{O}}] = \frac{1}{2}\eta^2 \operatorname{tr}(H\, \Sigma_\delta)$, so $P_{\mathcal{C}} = H$ exactly*.*

Proof. With $\theta^+ = \theta - \eta g$, expand $L(\theta^+ - \eta\delta) = L(\theta^+) - \eta\delta^\top H\theta^+ + \frac{1}{2}\eta^2\delta^\top H\delta$; the cross term is linear in δ and vanishes in expectation, leaving $\frac{1}{2}\eta^2 \operatorname{tr}(H\, \Sigma_\delta)$ with no higher-order remainder. □

Here $P_{\mathcal{C}} = H$ *exactly*: the loss curvature *is* the read operator, with no second-order remainder. Reconstruction ($\operatorname{tr}\Sigma_\delta$) is the flat-loss case $H = I$; when the loss is curved, a gradient code good in flat directions and bad in sharp ones is downstream-worse than its reconstruction admits.

Remark (Retrieval). For score-ranking consumers $s(q, x_i) = q^\top x_i$, the fraction of inverted pairs is, to leading order for smooth rank statistics, monotone in $\operatorname{tr}(P_{\mathcal{C}}\bar{\Sigma}_\delta)/\operatorname{tr}(P_{\mathcal{C}}\bar{\Sigma}_0)$ with $P_{\mathcal{C}} = \mathbb{E}[qq^\top]$ and $\bar{\Sigma}$ the across-item covariance—the same read operator, now a subspace projector when the query distribution is concentrated. (Rank statistics are non-smooth; we treat this instance empirically, §4.)

4 Distortion: the consumer is the metric (GO-2)

Proposition 2 predicts that at *matched reconstruction* the downstream ranking of two codes is set by $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$, hence by the query distribution. We test it under sealed registration.

Dissociation. Two 4-bit codes at an audited matched budget (spread 0.19 bit) with *identical* reconstruction (0.0938 vs 0.0934): the whole-vector code is 2.53× worse on softmax-KL (12/12 seeds); a wrong-quotient control is 21× worse at $\sim 5\times$ the reconstruction. Reconstruction is not the controlling quantity (`[demonstrated]`).

Flip. Holding reconstruction fixed and changing only the query bank, the ranking *inverts*: isotropic ⇒ code a better (2.5×, 12/12); subspace ⇒ code b better (12/12). The directly computed $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$ tracks the sign on every seed in both consumers; reconstruction, a consumer-blind scalar, cannot (rank correlation 0.40 vs 1.00). This is Proposition 2 with Q swapped, observed (`[demonstrated]`).

Replication and out-of-sample. The flip replicates on retrieval ranking (different representation, set-valued consumer): identical reconstruction (0.0964), ranking flips with the read subspace (log-advantage −4.70 vs +4.65, 12/12), $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$ tracking, reconstruction blind (`[replicated]`). It generalizes to optimization with $P_{\mathcal{C}} = H$ (Proposition 3): at matched reconstruction the curvature-metric distortion flips between two Hessians (12/12), tracked by $\operatorname{tr}(H\Sigma_\delta)$, reconstruction uninformative (−0.20)—and under $H = I$ reconstruction tracks *perfectly* (1.00), the derived $P_{\mathcal{C}} = I$ corner, observed (`[predicted]`, out-of-sample). One derivation; three read operators.

5 Identifiability: the metric is recoverable (GO-1)

Proposition 1 is useful only if $P_{\mathcal{C}}$ is knowable before compressing.

Proposition 4 (Recovery of the read subspace). *Let $\mathcal{C}(x) = f(Wx)$ with $W \in \mathbb{R}^{r\times D}$ of rank r and f coordinatewise. Then $J(x)^\top J(x) = W^\top \mathrm{diag}(f'(Wx)^2)\, W$ and $\mathbb{E}_x[J^\top J] = W^\top \mathbb{E}[\mathrm{diag}(f'^2)]\, W$ has rank r with column space equal to* row(W). *Hence the top-r eigenvectors of $\mathbb{E}_x[J^\top J]$ span the read subspace, and a finite-difference estimate $\hat{J}u \approx (\mathcal{C}(x{+}hu) - \mathcal{C}(x{-}hu))/2h$ recovers it from consumer queries alone.*

We built such a probe—receiving only the callable $\mathcal{C}(\cdot)$ and the dimensions, never W, the codes, or $d_{\mathcal{O}}$ (blinding enforced in code). It recovers the planted read subspace at overlap 0.936 against a 0.059 chance baseline (16×), and the *blind* $\mathrm{tr}(\hat{P}_{\mathcal{C}}\Sigma_\delta)$ predicts the flip of §4 on 12/12 seeds in both consumers with rank correlation 1.00—identical to ground truth—while reconstruction cannot (0.40) (`[predicted]`). Composed: from the consumer alone one predicts which code wins, before compressing anything.

6 Capacity: a vacuity threshold, derived (GO-3)

Observation has a reliability limit, and it too is computable in advance.

Proposition 5 (Retrieval vacuity threshold). *For single-stage top-1 retrieval with N–1 conditionally-exchangeable distractor scores standardized to mean 0, variance 1, success requires the standardized true margin to exceed $M_{N-1} = \max$ of the distractors, and*

$$\mu_{\mathrm{crit}} := \mathbb{E}[M_{N-1}] = \sqrt{2\ln(N{-}1)} - \frac{\ln\ln(N{-}1) + \ln 4\pi}{2\sqrt{2\ln(N{-}1)}} + o\Big(\tfrac{1}{\sqrt{\ln N}}\Big).$$

Writing the margin certificate $\hat{\mu} = (\overline{true} - \overline{distr})/\mathrm{std}$ and $\rho = \hat{\mu}/\mu_{\mathrm{crit}}$, recall crosses $\frac{1}{2}$ at $\rho \approx 1$; the certificate is vacuous *for $\rho < 1$.*

Derivation. Standardized i.i.d. Gaussian maxima obey the Fisher–Tippett law with Gumbel limit and centering/scaling $a_N = \sqrt{2\ln(N{-}1)}$, $b_N = 1/a_N$; $\mathbb{E}[M_{N-1}] = a_N - b_N(\ln\ln(N{-}1) + \ln 4\pi)/2 + o(b_N)$. Top-1 success is true $> M_{N-1}$; averaging the standardized margin gives $\hat{\mu}$, so the 0.5-recall crossing is at $\hat{\mu} \approx \mathbb{E}[M_{N-1}] = \mu_{\mathrm{crit}}$. □

Across 14 structurally distinct corpora the derived—not fitted—threshold locates death: the recall-0.5 crossing sits at $\rho = 0.948$ (within 6% of 1); the certificate orders corpora by recall at rank correlation 0.991, beating the un-normalized margin (0.873), with a finite-width transition (`[demonstrated]`). Capacity says, from a computable quantity, *where* the observation channel dies before any retrieval is run.

7 Rate: the verdict belongs to the budget (GO-4)

The read operator, capacity, and distortion are stated at a fixed observer budget; the budget is itself a rate, and it can invert the verdict. The mechanism is spectral. An eigenfunction φ_k of a Laplace–Beltrami operator with eigenvalue λ_k oscillates on the *wavelength* $\lambda_k^{-1/2}$; Weyl's law $N(\lambda) \sim c\,\mathrm{Vol}(M)\,\lambda^{d/2}$ says the count of modes below a fixed scale grows with resolution. Geodesic distance is preserved by modes whose wavelength *exceeds* the geodesic scale of interest (small λ); modes with λ above a threshold τ oscillate *below* that scale and encode sub-geodesic detail.

Lemma 1 (Budget inversion)**.** *Fix a low-mode budget m (a fixed low-λ band). As the sample N grows, the graph operator converges to the manifold operator and the fixed band* sharpens, *so geodesic fidelity is nondecreasing. Hold instead the resolved scale τ fixed (spectral-gap matching), so $m = N(\tau)$ grows with N: the newly resolved modes near τ have wavelength approaching the sampling scale and, for a non-ideal (finite-sample, noisy) substrate, inject sub-geodesic detail, so geodesic fidelity is nonincreasing. The two budgets coincide at the coarsest N and diverge in* opposite *directions thereafter.*

We test Lemma 1 on real 768-d book-paragraph embedding manifolds (32,000 points, three seeds). Fixed budget $m = 10$: geometry preservation rises with N (trend $+0.4, +1.0, +1.0$). Spectral-gap-matched, $m : 10 \to 121/126/159$: geometry *collapses* (trend -1.0 every seed), the added high modes injecting sub-wavelength detail; the budgets agree at the smallest N (gap ≤ 0.002) (`[replicated]`). Opposite verdicts about the same data, set by the observer's rate: "clean geometry" is a relation between observer resolution and substrate scale, not a substrate invariant.

8 A consumer-relative rate–distortion function

The four faces assemble into one object. For a Gaussian source and the consumer-projected distortion of Proposition 1:

Proposition 6 (Consumer-relative Gaussian rate–distortion)**.** *Let $x \sim \mathcal{N}(0, \Sigma_x)$ and constrain* $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta) \leq D$. *In the basis diagonalizing $P_{\mathcal{C}}^{1/2}\Sigma_x P_{\mathcal{C}}^{1/2}$ with eigenvalues γ_i, the minimum rate is*

$$R_{\mathcal{C}}(D) = \sum_i \tfrac{1}{2}\log_2^+ \frac{\gamma_i}{\theta}, \qquad \sum_i \min(\gamma_i, \theta) = D,$$

a reverse water-filling on the $P_{\mathcal{C}}$-weighted source spectrum. *For $P_{\mathcal{C}} = I$ this is Shannon's Gaussian rate–distortion function [2], and the estimation dual is Gauss–Markov least squares [3].*

Derivation. $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta) = \operatorname{tr}(P_{\mathcal{C}}^{1/2}\Sigma_\delta P_{\mathcal{C}}^{1/2})$; change variables $y = P_{\mathcal{C}}^{1/2}x$ so the distortion is ordinary MSE on y with source covariance $P_{\mathcal{C}}^{1/2}\Sigma_x P_{\mathcal{C}}^{1/2}$. Shannon's Gaussian water-filling applies to y, giving the stated allocation; the transform is invertible off the null space of $P_{\mathcal{C}}$, on which the consumer is blind and no rate is spent. □

Proposition 6 makes the placement exact: *classical rate–distortion is Observation Theory at* $P_{\mathcal{C}} = I$. Bits should water-fill on the directions the consumer reads *and* the source varies (γ_i large), and spend nothing on $\ker P_{\mathcal{C}}$—which is precisely why, empirically, a reconstruction-optimal code ($P_{\mathcal{C}} = I$ allocation) is beaten by a consumer-matched one whenever $P_{\mathcal{C}} \neq I$, and why the winner *flips* with $P_{\mathcal{C}}$ (§4). The identity-curvature control of §4 is the clean confirmation: only at $P_{\mathcal{C}} = I$ does reconstruction become the correct objective, and there it becomes it perfectly.

A more general Shannon: lossless to the consumer below the source entropy

Proposition 6 departs from Shannon in a way worth stating sharply. Shannon source coding obeys $R(0) = H(X)$: lossless representation costs the full source entropy. Observation Theory does not.

Proposition 7 (Consumer-lossless floor)**.** *Let Π project orthogonally onto* $\operatorname{range}(P_{\mathcal{C}})$*, rank $r \leq D$. There exist codes with $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta) = 0$—zero downstream distortion—at rate*

$$R_{\mathcal{C}}(0) = H(\Pi X) \;\leq\; H(X), \qquad H(X) - R_{\mathcal{C}}(0) = H\big(X \mid \Pi X\big),$$

with equality iff $P_{\mathcal{C}} \succ 0$. The saving is the entropy of the consumer-invisible subspace $\ker P_{\mathcal{C}}$*, discarded at no downstream cost.*

Proof. $\mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta) = \mathrm{tr}(P_{\mathcal{C}}\,\Pi\Sigma_\delta\Pi)$ depends on δ only through $\Pi\delta$; a code that transmits ΠX losslessly and reconstructs $(\mathrm{I}-\Pi)X$ by its conditional mean has $\Pi\delta = 0$, hence $\mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta) = 0$, at rate $H(\Pi X)$. □

This is a genuinely more general statement than Shannon's: *one can be lossless to the consumer while compressing below the source entropy*, and the floor is the entropy of the *read subspace*, not of the source. It is what the flip experiments exhibit—a code spending zero bits on $\ker P_{\mathcal{C}}$ (the across-token mean the softmax cannot see, Prop. 2; the subspace the query bank never probes) loses nothing the consumer can detect, and beats a reconstruction-optimal code that wastes bits there. Shannon is the corner $\ker P_{\mathcal{C}} = \{0\}$, where nothing is free.

Placed against the literature, Observation Theory is the middle term of a chain: Shannon rate–distortion ($P_{\mathcal{C}} = I$) $\subset$ consumer-relative rate–distortion ($P_{\mathcal{C}}$ a rank-r *geometric, identifiable* operator, Props. 6, 7) $\subset$ the information bottleneck [5] (an arbitrary relevance variable Y, generally non-geometric and requiring the joint $p(X,Y)$). The middle term is the operational one: $P_{\mathcal{C}}$ is the *local Fisher metric* of the relevance $\mathcal{C}(X)$ (Prop. 1), it is PSD and water-fillable (Prop. 6), and—unlike the abstract bottleneck—it is *recoverable from the consumer alone* by probing (Prop. 4). A more general Shannon is thus not a slogan but a program: replace "represent the source" with "serve the reader," make the reader's metric a measurable geometric object, and recover source coding, the reliability limit of §6 (a consumer-relative channel bound), and least squares as the $P_{\mathcal{C}} = I$ shadow. What remains for the converse is a coding theorem: that $R_{\mathcal{C}}(D)$ is *achievable* for general (non-Gaussian) sources by a random-coding argument on the $P_{\mathcal{C}}$-tilted typical set—the natural next theorem of this program.

9 A boundary: the refuted quotient (GO-5)

Not every plausible face survives a sealed prediction. In the spectral setting the α=1 anisotropic normalization of a diffusion operator cancels sampling density and recovers density-free geometry [4]; the null space of $P_{\mathcal{C}}$ there is the density mode. We conjectured this *density quotient* likewise restores the invariant in a *non-spectral* domain (locally-scaled ADC retrieval). It does not. Across four prospective tests—direct affinity (two variants), product-quantization ADC on real embeddings, and non-spectral diffusion-distance retrieval—the quotient fails to produce a decisive, density-specific restoration. In the direct and ADC domains it *hurts*: the retrieval base is not a density-biased operator, so there is no density term to cancel, and in concentrated real data density is entangled with the invariant, so down-weighting dense points removes signal. In the diffusion domain the direction is right—α=1 is the optimal exponent, recovered non-spectrally by iterated transition-matrix powers—but the effect is under 2% and *not density-specific* (a uniform-density control gains as much, +0.015 vs +0.019). Per its sealed registration the conjecture is `[refuted]` (NEG-11). The sharpened statement: *the α=1 cancellation is a property of the diffusion operator and does not transfer to direct observation*—consistent with Proposition 6, which spends zero rate on $\ker P_{\mathcal{C}}$ only when $P_{\mathcal{C}}$ actually contains the density direction. The theory is stronger for the boundary.

10 Honest negatives

Eleven standing negatives bound the theory and are reported with the prominence of the positives. Reconstruction cosine is not a proxy for key quality (the motivating negative). The reconstruction-blind flip is observable only for reconstruction-*matched* codes: absent a tie one code Pareto-dominates and reconstruction is not falsified (NEG-10, a prospective miss locating a precondition, consistent with Proposition 6's water-filling). The projected-variance *trace* governs the discriminating comparison but is not a complete rank statistic over all codes (NEG-9). The density quotient (NEG-11, §9). Two are methodological: twice a substantive claim passed at 12/12 while an over-strict secondary gate we had written failed, and twice a degenerate substrate was caught only by a regime-coverage gate—reported because a theory that publishes only its wins cannot be believed about them.

11 Method: registration-first

Every empirical claim was governed by a dated, hashed registry entry committed—sealed—before the first measurement it governs; the commit is the binding timestamp. Bars were fixed ex ante; misses enter the ledger as misses; amendments only dated and logged before unblinding; prospective claims (§5,7,9) were blinded where feasible. The registry, ledger, harnesses, and result artifacts are public; the house rule is that the paper may not assert what the ledger cannot show. This is what lets a reader separate the four faces that held from the fifth that did not.

12 Discussion

Observation Theory asks a change of default. Where a pipeline minimizes reconstruction and hopes the task follows, the theory says: recover the consumer's read operator (§5), water-fill bits on $P_{\mathcal{C}}$ rather than on I (§8), respect the channel's vacuity threshold (§6), and remember the verdict is relative to the observation budget (§7). The practical target is compression, quantization, and retrieval, where reconstruction reigns and is, we have shown analytically and empirically, the wrong objective when $P_{\mathcal{C}} \neq I$. The theoretical content is a reframing: geometry, distortion, and reliability are relations between an object and an observer, and the observer is specifiable, recoverable, and—within its budget and channel—sovereign over the verdict. Open: the achievability and converse of $R_{\mathcal{C}}(D)$ beyond the Gaussian case; whether the flip is exactly its argmin migrating with $P_{\mathcal{C}}$; and the hardening of §5–4 on trained-model consumers. Four faces, one refutation, eleven negatives: a world to map, not a slogan.

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Reproducibility & provenance. All results resolve to sealed, before-run registry entries and committed artifacts in the companion repository. Computational experimentation and drafting were carried out with an AI research assistant under the registration-first protocol of §11.